

AU Campus Recruitment 2026

AI/ML Engineer – Individual Technical Assessment

Overview

As part of AU Campus Recruitment 2026, candidates shortlisted for the AI/ML Engineer role are required to complete an individual 24-hour technical assessment.

Each candidate must select any ONE of the two problem statements provided in the document and submit the solution within 24 hours from the time of assignment.

The assessment evaluates:

- Machine Learning fundamentals
- NLP / LLM integration
- Personalization logic
- Model reasoning and explainability
- System design clarity

No dataset will be provided. Candidates must use the allowed public datasets mentioned in the problem statement document or generate synthetic data with proper justification.

Submissions after the 24-hour deadline will not be considered.

Start : 5:30 PM (18-02-2026)

Deadline : 5:30 PM (19-02-2026)

Deliverables

Candidates must submit:

- Public GitHub Repository Link
- Clean and structured code
- Proper README with setup instructions

Technical Documentation (PDF)

- Approach explanation
- Dataset source
- Model architecture
- Assumptions and limitations
- Sample outputs

Short Demo Video (3–5 minutes)

- Working solution demonstration
- Explanation of model decisions

Submission Link

- Submit all deliverables through the following link:

<https://docs.google.com/forms/d/e/1FAIpQLScPihEqkfEeniVQPrQL7n-MRBS1o1kA0IzvOkRFD7XOc4vzCg/viewform?usp=publish-editor>

Problem Statement - 1

Build an AI-powered teaching assistant that focuses on the following two core capabilities:

1. Student Query Understanding
2. Adaptive Learning Path Recommendation

The system should simulate how an EdTech platform understands students and adapts learning accordingly.

Q1: Student Query Understanding (NLP + LLM)

Input: A student asks a question in natural language.

Example: "I don't understand backpropagation."

The system must classify the query into:

- **Intent type** (Explanation, Example, Doubt clarification, Revision)
- **Topic** (e.g., Optimization, Neural Networks)
- **Difficulty level** (Beginner / Intermediate / Advanced)

Expected Techniques:

- Sentence embeddings (BERT, MiniLM, SBERT, or equivalent)
- Topic classification using Transformer or classical ML models
- LLM prompt engineering for intent refinement

Expected Output Example:

```
{  
  "intent": "Explanation", "topic":  
    "Backpropagation",  
  "difficulty_level": "Intermediate"  
}
```

Q2: Adaptive Learning Path Recommendation

Based on a student's past performance and interactions, the system should recommend how the student should proceed next.

Inputs may include:

- Quiz scores
- Number of attempts
- Time spent on topics
- Previously asked questions

The system must recommend:

- Next topic to study
- Whether revision is required
- Difficulty adjustment (increase / decrease / same)

Expected Techniques:

- Reinforcement learning (basic policy-based methods)
- Rule-based + ML hybrid systems
- Simple knowledge tracing or clustering

Expected Output Example:

```
{  
  "next_topic": "Gradient Descent Learning Rate", "action": "Revision",
```

```
"difficulty_adjustment": "Decrease"
}
```

Dataset Usage Guidelines (Mandatory)

No dataset will be provided by the organizers.

Participants must use one or more of the following publicly available datasets or generate synthetic data.

Allowed Public Dataset Links:

- SQuAD – <https://rajpurkar.github.io/SQuAD-explorer/>
- SciQ – <https://allenai.org/data/sciq>
- AI2 ARC Dataset – <https://allenai.org/data/arc>
- ASSISTments Dataset – <https://sites.google.com/site/assistmentsdata/>
- EdNet Dataset – <https://github.com/riid/ednet>

Synthetic Data Guidelines:

- Clearly explain how the data was generated
- State assumptions made
- Mention limitations
- Dataset size must not exceed 50 MB

What We Expect From Teams

Judges will focus on:

- Correct intent classification (not random or rule-only)
- Meaningful use of embeddings or ML models
- Logical adaptation of learning paths
- Clear explanation of models and decisions

UI quality is not a priority.

Evaluation Criteria

Area	Weight	What We Check
Problem Decomposition	15%	Clear ML vs LLM separation
ML Algorithm Depth	25%	Intent classification, clustering, RL logic
LLM Usage Quality	20%	Prompt design and consistency
Personalization Logic	15%	Adaptive, non-generic outputs
System Design	15%	Modular and explainable design
Innovation	10%	Novel approaches or insights

Time Constraint

- Total duration: 24 hours
- Internet access is allowed

- Pre-trained models may be used with proper explanation